

Name _____

Class: 17S _____

Reg Number: _____



MERIDIAN JUNIOR COLLEGE
JC 2 Preliminary Examination
Higher 2

Chemistry

9729/03

Paper 3 Free Response

13 September 2018

2 hour

Additional Materials: *Data Booklet*
 Writing Paper

INSTRUCTIONS TO CANDIDATES

Write your name, class and register number in the spaces provided at the top of this page.

Answer all questions in Section A and one question from Section B.

Begin each question on a fresh page of writing paper.

Fasten the writing papers behind the given **Cover Page for Questions 1 & 2** and **Cover Page for Questions 3 & 4 or 5** respectively.

Hand in Questions **1 & 2** and **3 & 4 or 5** separately.

You are advised to spend about 30 minutes per question only.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question.

You are reminded of the need for good English and clear presentation in your answers.

This document consists of **17** printed pages (including this cover page).

Section A

Answer **all** questions in this section.

- 1(a)** Explain the variation in melting point of the oxides of the elements in Period 3 (sodium to sulfur), using oxides of three different structure as examples.

Indicate clearly the type of structure and bonding present in the examples you have selected.

[3]

- (b)** With the aid of the *Data Booklet*, explain the difference in observed melting points of NaCl (801 °C) and MgO (2852 °C).

[2]

- (c)** Instead of melting like chlorides or oxides, Group 2 nitrates decompose upon heating.

- (i)** Write an equation, including state symbols, for the thermal decomposition of magnesium nitrate, $\text{Mg}(\text{NO}_3)_2$.

[1]

- (ii)** Predict, with reasoning, whether barium nitrate, $\text{Ba}(\text{NO}_3)_2$, is more or less thermally stable compared with $\text{Mg}(\text{NO}_3)_2$.

[2]

- (d)** Chlorides of Period 3 elements dissolve in water to form solutions of differing pH values.

- (i)** Sketch a graph to illustrate how the pH of the resultant solutions vary across the period (from sodium to phosphorus).

[1]

- (ii)** Write an equation to illustrate the reaction of PCl_5 with water.

[1]

- (iii)** Lithium is a Group 1 element. Its chloride, LiCl however behaves similarly to that of a chloride of a Group 2 element when dissolved in water.

Suggest a reason why this would be so.

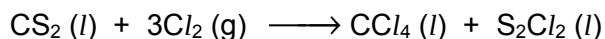
[1]

- (e)** Disulfur dichloride, S_2Cl_2 is one of three chlorides of sulfur. It is a liquid at room temperature with a pungent odour. It has a structure in which the central sulfur atoms are bonded by a single covalent bond.

Draw clear diagrams of two possible molecular arrangements of S_2Cl_2 , showing clearly the electron pairs around the sulfur atoms. Apply the principles of the VSEPR theory to suggest which arrangement would result in a molecule which could be more stable.

[3]

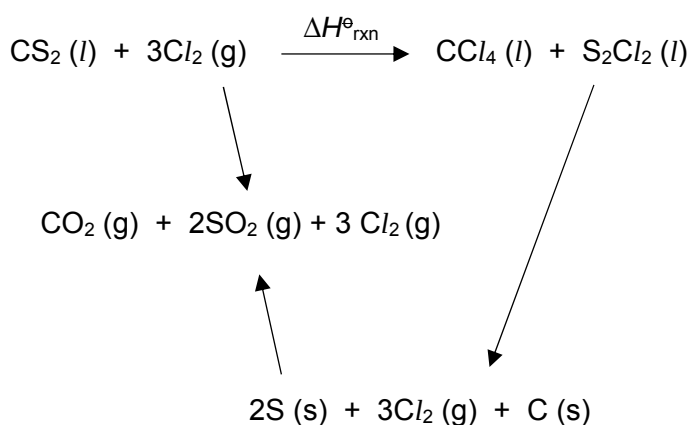
- (f) CCl_4 , an important commercial solvent is prepared by the reaction of Cl_2 gas with a sulfur-containing carbon compound. S_2Cl_2 is produced as a by-product. The reaction for the production is shown below.



- (i) Using appropriate data in **Table 1.2** and the energy cycle provided, determine the enthalpy change, $\Delta H^\circ_{\text{rxn}}$ for the above reaction.

Table 1.2

	$\Delta H^\circ / \text{kJ mol}^{-1}$
Standard enthalpy change of formation of $\text{CCl}_4(l)$	-135.4
Standard enthalpy change of combustion of C (s)	-393.5
Standard enthalpy change of formation of $\text{S}_2\text{Cl}_2(l)$	-58.2
Standard enthalpy change of formation of $\text{SO}_2(g)$	-296.8
$\text{CS}_2(l) + 3\text{O}_2(g) \longrightarrow \text{CO}_2(g) + 2\text{SO}_2(g)$	-1077.0
$\text{SO}_2(g) + \text{Cl}_2(g) \longrightarrow \text{SO}_2\text{Cl}_2(l)$	+97.3



- (ii) CCl_4 could be reacted with limited oxygen according to the following reaction.



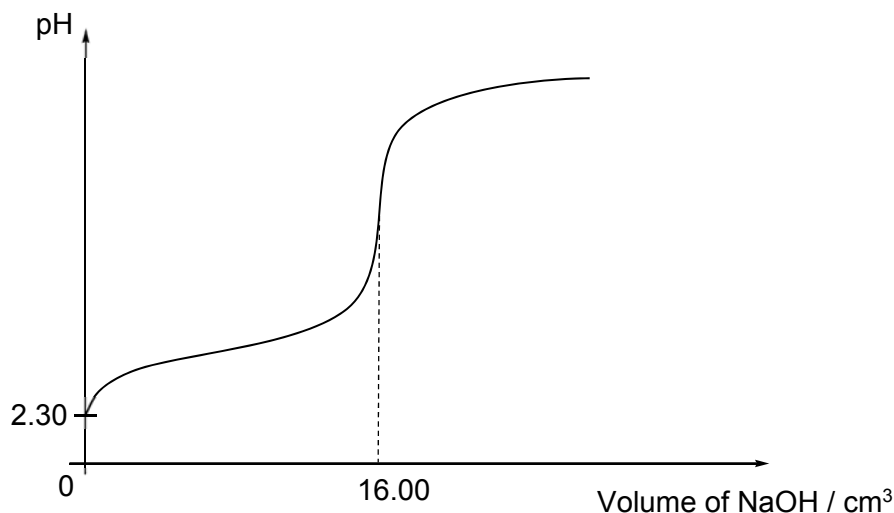
Predict the sign of entropy change, ΔS° for this reaction and explain whether the reaction would be spontaneous at all temperatures.

[2]

[Total: 18]

2 In analytical chemistry, both quantitative and qualitative information about chemical reactions are established through the use of various chemical methods.

- (a) Glycolic acid, HOCH_2COOH is a weak acid used in many cosmetic skin-care products. When a student titrated 25.0 cm^3 of a glycolic acid against $0.250 \text{ mol dm}^{-3}$ NaOH, the following titration curve was obtained.



- (i) Suggest an appropriate indicator for the titration, giving a reason for your choice.

[1]

- (ii) Use the above titration data to calculate the K_a value of glycolic acid.

[2]

- (b) The pH levels of skin-care products need to be regulated to prevent degradation of the active ingredients. A buffer solution containing ethanoic acid and its sodium salt, sodium ethanoate can be added to control the pH. The $\text{p}K_a$ of ethanoic acid is 4.76.

- (i) With the aid of a balanced equation, briefly explain how a solution of ethanoic acid and sodium ethanoate can maintain a fairly constant pH when a small amount of acid is added to this solution.

[1]

- (ii) Determine the resulting pH of a 100.0 cm^3 solution containing 0.10 mol dm^{-3} of ethanoic acid and 0.10 mol dm^{-3} of sodium ethanoate when $5.0 \times 10^{-4} \text{ mol}$ of H_2SO_4 is added to this solution.

[2]

- (c) Another student performed a series of thermometric experiments to determine the enthalpy change of neutralisation involving several acids and sodium hydroxide. Her experimental results are summarised in the table below.

Acid	Enthalpy change of neutralisation, $\Delta H_n^\circ / \text{kJ mol}^{-1}$
hydrochloric acid, HCl	– 57.2
hydrofluoric acid, HF	– 68.6
glycolic acid, HOCH_2COOH	– 53.2

- (i) Explain why the neutralisation reaction involving glycolic acid is less exothermic than that involving hydrochloric acid.

[1]

Deducing the strength of acids solely from the enthalpy change of neutralisation may not be appropriate as there are exceptions. For instance, the neutralisation reaction between HF , a weak acid and NaOH is significantly more exothermic than that of HCl and NaOH .

- (ii) Suggest why the neutralisation reaction between HF and NaOH is unexpectedly more exothermic.

[1]

- (d) Argentometric titration can be used to determine the amount of sodium chloride present in skin-care products by titrating the sample against silver nitrate, AgNO_3 . A suitable indicator is one that forms a precipitate of a different colour after all the chloride ions have reacted.

The following are relevant K_{sp} values of some sparingly soluble silver compounds.

	K_{sp}
AgCl	$1.8 \times 10^{-10} \text{ mol}^2 \text{ dm}^{-6}$
AgI	$7.7 \times 10^{-17} \text{ mol}^2 \text{ dm}^{-6}$
Ag_3PO_4	$1.3 \times 10^{-20} \text{ mol}^4 \text{ dm}^{-12}$

- (i) Write an expression for the solubility product, K_{sp} of Ag_3PO_4 .
- (ii) Given that Ag_3PO_4 precipitate is yellow in appearance, by means of suitable calculations, suggest whether potassium phosphate, K_3PO_4 can be used as an indicator in the argentometric titration for the determination of chloride content.
- (iii) With the aid of appropriate equations, explain why silver chloride, AgCl is soluble in excess aqueous ammonia.
- (iv) Hence, briefly explain why silver iodide, AgI remains insoluble even in concentrated ammonia solution.

[1]

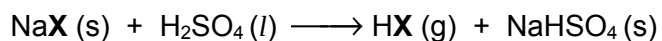
[2]

[3]

[1]

- (e) The distinctive reactions of solid halides with hot concentrated sulfuric acid also provide useful basis for further analysis.

These halides react with hot concentrated sulfuric acid to form white fumes of hydrogen halides.



where **X** = Cl, Br or I

However, only HBr and HI formed will further react with concentrated H_2SO_4 . The main products for these reactions are recorded in the table below.

HCl	HBr	HI
no further reaction	red–brown gas	violet fumes
	acidic gas that decolourises acidified purple KMnO_4	pungent H_2S gas

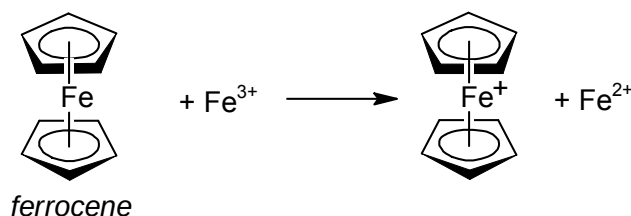
- (i) Suggest an identity for both the red–brown gas and the acidic gas formed when HBr further reacts with concentrated H_2SO_4 . [1]
- (ii) Hence, write an equation for the reaction between HBr and concentrated H_2SO_4 . [1]
- (iii) By considering the reactions of HCl and HI with concentrated sulfuric acid and the change in oxidation number of sulfur, explain which halide ion is a stronger reducing agent. [2]

[Total: 19]

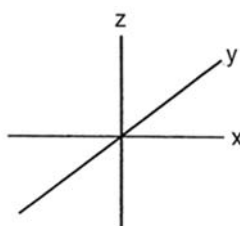
- 3 Transition elements show typical properties that distinguish them from s-block elements such as calcium. The following table gives data about some physical properties of the elements calcium, iron, copper and cadmium.

property	calcium	iron	copper	cadmium
relative atomic mass	40.1	55.8	63.5	112.4
atomic radius (metallic) / nm	0.197	0.126	0.128	0.151
ionic radius (2+) / nm	0.099	0.076	0.069	0.097
melting point / °C	839	1535	1085	322
density / g cm ⁻³	1.54	7.86	8.92	8.65

- (a) (i) Explain why the atomic radii of iron and copper are similar to each other. [2]
- (ii) The melting point of iron is significantly higher than that of calcium and cadmium. Explain this using relevant data from the table and in terms of the structure and bonding in each metal. [3]
- (b) Ferrocene, $\text{Fe}(\text{C}_5\text{H}_5)_2$, is an orange organometallic solid. In this complex, C_5H_5^- is the ligand and it donates π electrons from the ring to the vacant $3d$ orbitals of Fe. A chemist suggested that ferrocene would react with aqueous iron(III) ions to form iron(II) ions as shown below.



- (i) The chemist found the reduction potential of ferrocene, $E^\ominus_{\text{Fe}(\text{C}_5\text{H}_5)_2^+ / \text{Fe}(\text{C}_5\text{H}_5)_2}$ to be +0.63V. Predict if the reaction between ferrocene and iron(III) ions will occur. Calculate $\Delta G^\ominus$ for this reaction. [2]
- (ii) Iron(II) ion forms octahedral complexes, the orbitals are split into two energy levels. Using the Cartesian axes shown below, draw separate labelled diagrams of
- one of the d orbitals at the **lower** energy level in an octahedral complex,
 - one of the d orbitals at the **higher** energy level in an octahedral complex.



[2]

- (iii) Using your diagrams in (b)(ii), explain why the d subshell of a transition metal ion is split into two energy levels in an octahedral complex.

[2]

Most transition element complexes are coloured.

- (iv) Explain why ferrocene is orange in colour.

[2]

- (v) Aqueous Fe^{2+} ion is green in colour, suggest and explain if water causes a larger or smaller split between the two groups of $3d$ orbitals as compared to C_5H_5^- .

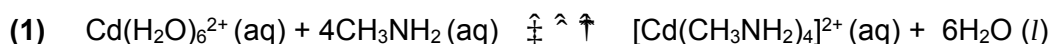
[2]

- (vi) The reaction between $\text{S}_2\text{O}_8^{2-}$ and I^- is slow in the absence of catalyst. $\text{Fe}^{2+}(\text{aq})$ can be used as catalyst for the reaction between $\text{S}_2\text{O}_8^{2-}$ and I^- .

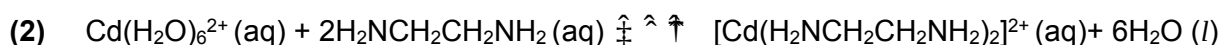
Explain with the aid of equations how it works.

[3]

- (c) Cadmium ions form complexes with primary amines and with 1,2-diaminoethane.



$$K_{\text{stab}} = 3.6 \times 10^6$$



$$K_{\text{stab}} = 4.2 \times 10^{10}$$

K_{stab} is an equilibrium constant that accounts for the formation of a complex. It is a measure of the strength of the interactions between the ligands and the metal centre that come together to form the complex.

The values for ΔH° and ΔG° for equilibria (1) and (2), and the value of ΔS° for equilibrium (1), are given in the table below. All values are determined at a temperature of 298 K.

Equilibrium	$\Delta H^\circ / \text{kJ mol}^{-1}$	$\Delta G^\circ / \text{kJ mol}^{-1}$	$\Delta S^\circ / \text{J mol}^{-1} \text{K}^{-1}$
(1)	- 57.3	- 37.4	- 66.8
(2)	- 56.5	- 60.7	-

- (i) Suggest a reason why the ΔH° values for the two equilibria are very similar.

[1]

- (ii) Calculate the standard entropy change of reaction, ΔS° in equilibrium (2).

[2]

(iii) Suggest a reason for the difference between the $\Delta S^\ominus$ you have calculated for equilibrium (2) and that for equilibrium (1) given in the table.

[1]

(iv) Which of the two complexes is more stable? Give a reason for your answer.

[1]

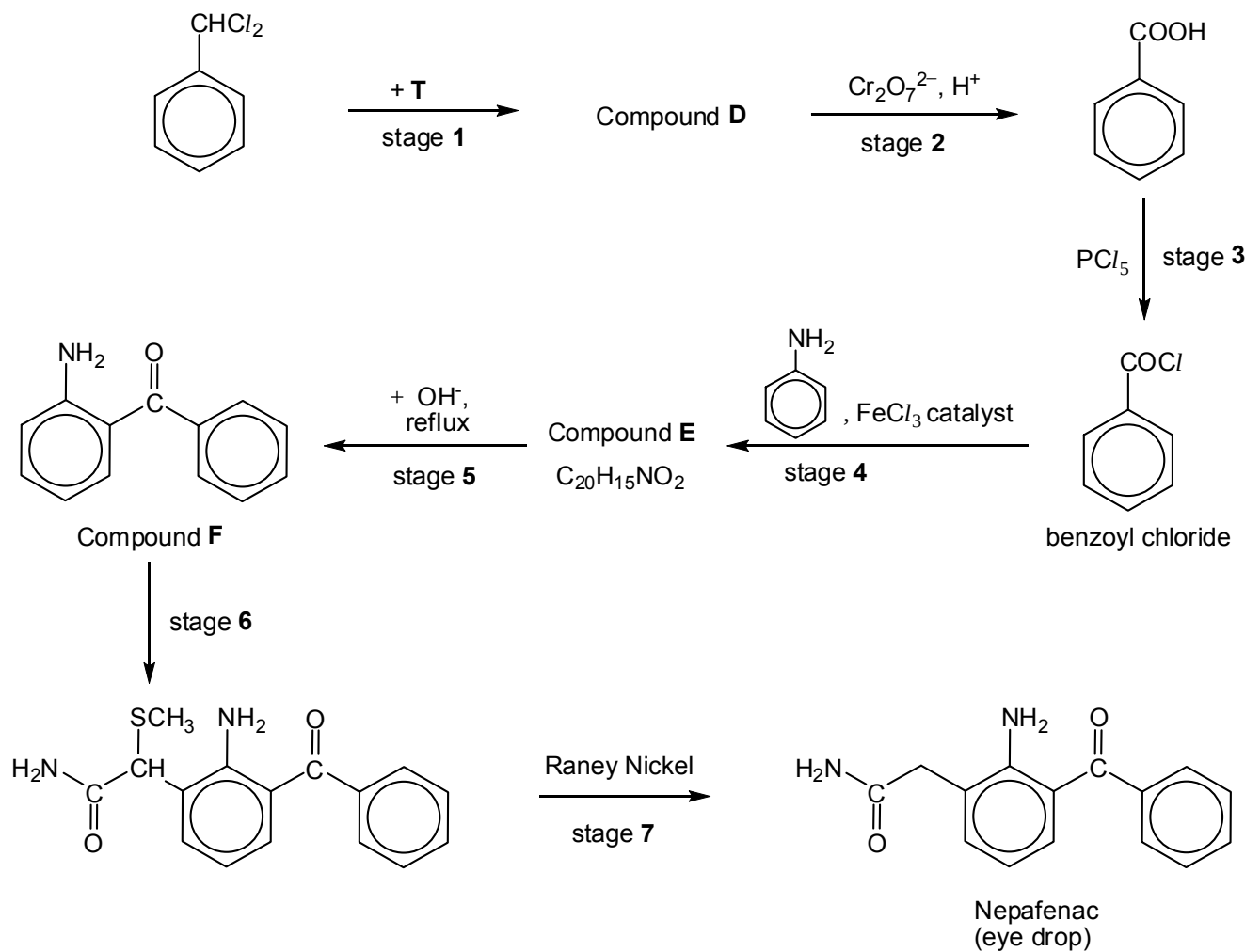
[Total: 23]

Section B

Answer **one** question from this section.

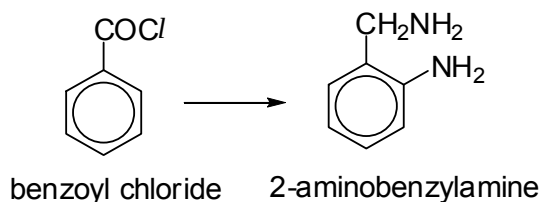
- 4(a)** *Nepafenac* is a prescription eye drop that is used to treat pain and inflammation associated with eye surgery.

The following scheme shows a synthesis of *Nepafenac*.



- (i) State the type of reaction occurring in stage 3 and stage 7. [2]
- (ii) Draw the structure of compound **D**. Suggest the reagent and conditions required in stage 1. [2]
- (iii) In stage 4, two reactions are involved. Draw the structure of compound **E** and state the reactions that occurred. [2]

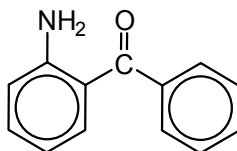
- (iv) Benzoyl chloride can be converted into 2-aminobenzylamine, an important starting material in the synthesis of quinazoline, a source of useful pharmacophores for new drug development.



Suggest a 4-step synthesis of 2-aminobenzylamine from benzoyl chloride. You should state the reagents and conditions needed for each step, and show clearly the structures of any intermediate compounds.

[4]

- (v) A student made a statement regarding a reaction with compound **F**: “Cyanide, CN^- , is acting as a **Bronsted–Lowry base** to attack the carbonyl group of compound **F** to form a cyanohydrin.”

Compound **F**

Based on your understanding of a *Bronsted–Lowry* base, explain why the student's statement is wrong.

[1]

- (vi) Data about phenylamine and diphenylamine are given in the table below.

Compound	 phenylamine	 diphenylamine
pK_b value	9.39	13.21

Give **two** reasons, why pK_b of phenylamine is lower than that of diphenylamine.

[2]

- (b) Compound **J**, $C_{10}H_8O_3$, does not decolourise bromine water nor give effervescence with magnesium strip. **J** gives a silver mirror with Tollens' reagent but does not give a precipitate with hot alkaline Cu^{2+} solution.

On prolonged heating with acidified $KMnO_4$, **J** gives benzene-1,2,3-tricarboxylic acid as the only organic product. When heated with dilute H_2SO_4 , **J** forms compound **K**, $C_{10}H_{10}O_4$. **K** gives a yellow precipitate with alkaline aqueous iodine.

J reacts with $LiAlH_4$ to form compound **L** (with an internal plane of symmetry), $C_{10}H_{14}O_3$. However, **J** forms compound **M**, $C_{10}H_{10}O_3$ with $NaBH_4$.

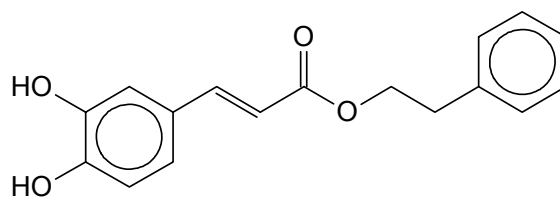
All compounds are optically active but only **K** reacts with sodium carbonate.

Suggest the structure for **J**, **K**, **L** and **M** and explain the reactions described.

[7]

[Total: 20]

- 5 Caffeic acid phenethyl ester (CAPE) is a natural occurring compound, found in propolis from honeybee hives. It is currently being studied for its anti-carcinogenic and anti-inflammatory properties. It is the ester of caffeic acid and 2-phenylethanol.



caffeic acid phenethyl ester (CAPE)

- (a) Till date, there has been two well-established reaction schemes to synthesise CAPE. Study **Fig. 2.1** and **Fig. 2.2** carefully and answer the questions that follow.

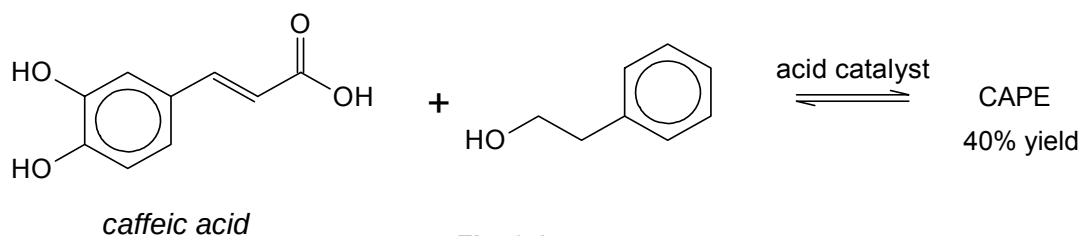


Fig. 2.1

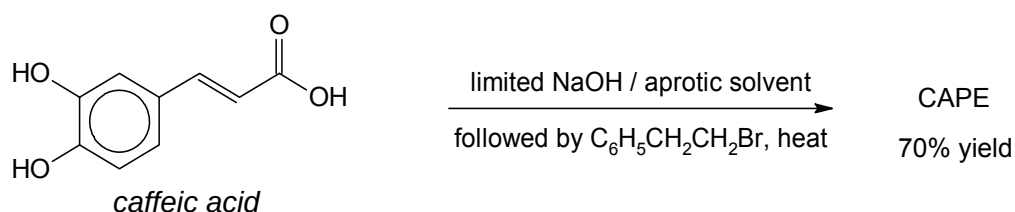


Fig. 2.2

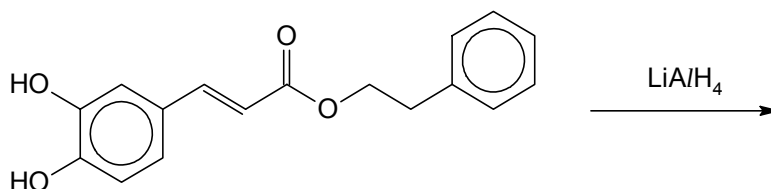
- (i) State the type of reaction occurring in each of the above synthesis method. [2]
- (ii) To improve the yield of CAPE in **Fig 2.1**, a student suggested using the acyl chloride derivative of caffeic acid instead. Comment on the validity of his suggestion. [2]
- (iii) Explain why NaOH has to be added in limited amount rather than in large excess in **Fig 2.2**. [1]
- (iv) A protic solvent is one that has labile proton(s) and readily donates its proton(s). Conversely an aprotic solvent does not have any labile proton. Suggest why is it important to use an aprotic solvent in **Fig 2.2**. [1]

- (v) The reported yield in **Fig. 2.2** is only 70%, as there are other by-products. Draw the skeletal formula of one possible by-product, assuming that caffeic acid reacts with $\text{C}_6\text{H}_5\text{CH}_2\text{CH}_2\text{Br}$ in a 1:1 ratio.

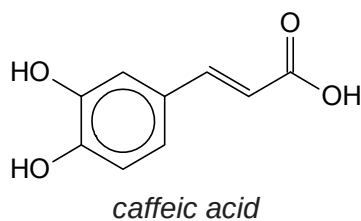
[1]

- (vi) Draw the structure of the resultant product(s) when CAPE is reacted with LiAlH_4 .

[1]



- (b) The structure of *caffeic acid* is shown below.

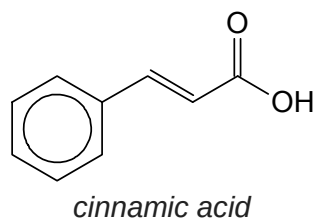


- (i) *Caffeic acid* contains two organic functional groups with pK_a values lower than that of an alcohol. Identify these functional groups.

Compare and account for the relative pK_a values of these two organic groups.

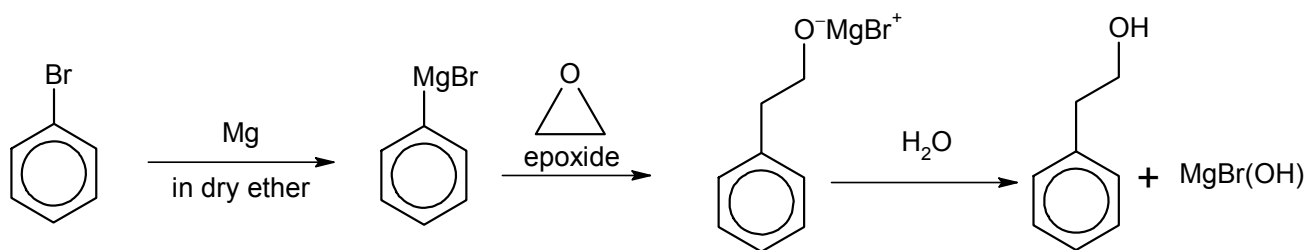
[3]

- (ii) Suggest a chemical test that can be carried out to distinguish between a sample of *caffeic acid* and *cinnamic acid*.



[2]

(c) The synthesis of 2-phenylethanol from bromobenzene is shown below.



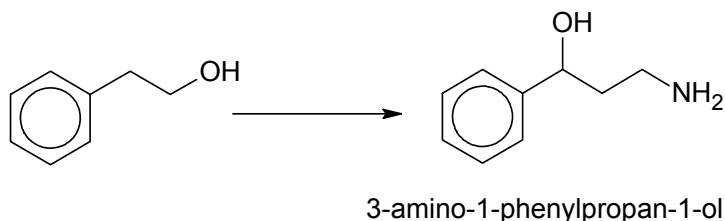
In the above synthesis, a Grignard reagent, $\text{C}_6\text{H}_5\text{MgBr}$ is generated. Once generated, the Grignard reagent behaves as a nucleophile. The C_6H_5 in $\text{C}_6\text{H}_5\text{MgBr}$ behaves like an anion, C_6H_5^- and is a strong Lewis base. The Grignard reagent is useful for the formation of carbon-carbon bonds.

The reaction between the Grignard reagent, $\text{C}_6\text{H}_5\text{MgBr}$ and the epoxide, $\text{C}_2\text{H}_4\text{O}$ is a bimolecular reaction and the intermediate formed will subsequently react with water to form 2-phenylethanol.

From the information provided, outline the mechanism to account for this reaction. In your answer, show any relevant charges, lone pairs of electrons and movement of electrons by curly arrows.

[2]

(d) Suggest a 4-step synthesis of 3-amino-1-phenylpropan-1-ol from 2-phenylethanol.



The synthesis involves an intermediate compound that decolourises aqueous bromine in the dark. You should state the reagents and conditions needed for each step, and show clearly the structures of any intermediate compounds.

[3]

(e) A 0.460 g sample of gaseous methanoic acid, HCOOH was found to only occupy a volume of 173 cm^3 at a temperature of 160°C and a pressure of $1.04 \times 10^5 \text{ Pa}$.

Using the ideal gas equation and the above information, determine the molar mass of this gaseous sample. Hence, draw a possible structure at this temperature.

[2]

[Total: 20]

END OF PAPER

Cover Page for P3 Q1 & 2

Name _____

Class: 17S _____

Reg Number: _____



MERIDIAN JUNIOR COLLEGE
JC 2 Preliminary Examination
Higher 2

13 September 2018

INSTRUCTION TO CANDIDATES

Write your name, class and register number in the spaces provided at the top of this page.

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Answer **only ONE** question in Section **B**.

Begin each question on a ***fresh page*** of writing paper.

Fasten your answers for Q1 & 2 behind this Cover Page.

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You are reminded the need for good English and clear presentation in your answers.

Examiner's Use		
Paper 3	Q1	/ 18
	Q2	/ 19

Cover Page for P3 Q3 & 4 or 5

Name _____ Class: 17S _____ Reg Number: _____



MERIDIAN JUNIOR COLLEGE
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Higher 2

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Examiner's Use		
Paper 3	Q3	/ 23
	Q4	/ 20
	Q5	/ 20

